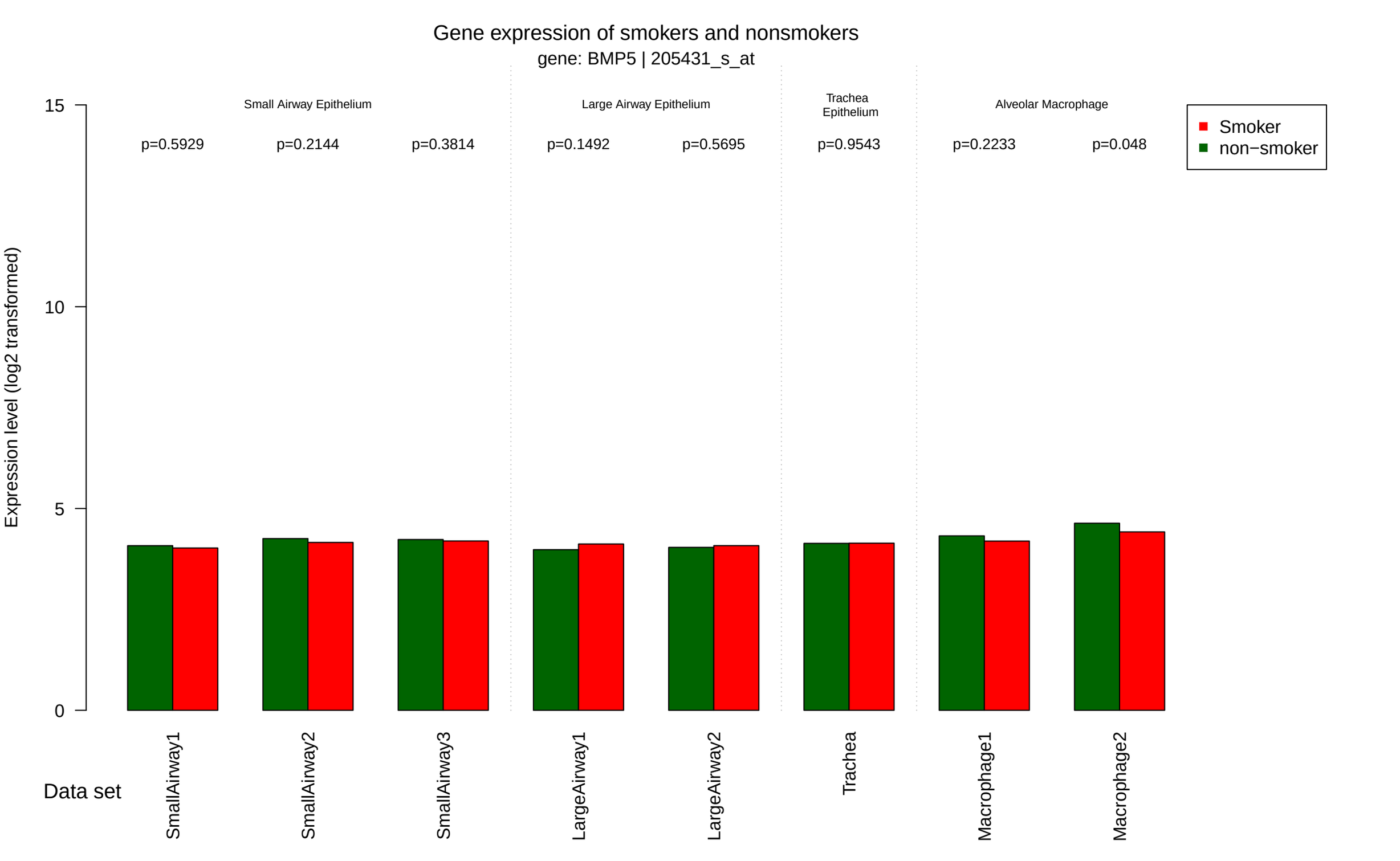
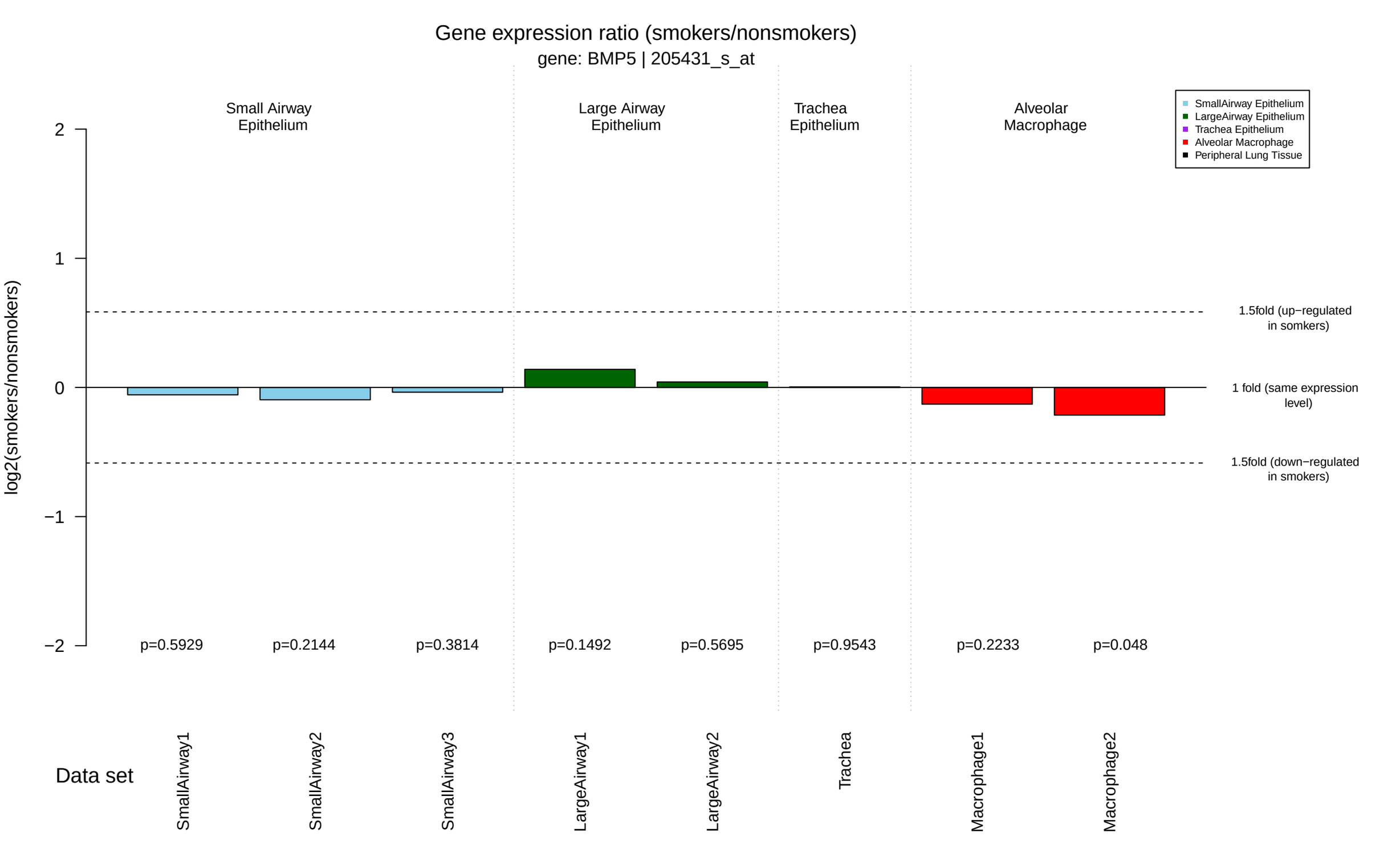
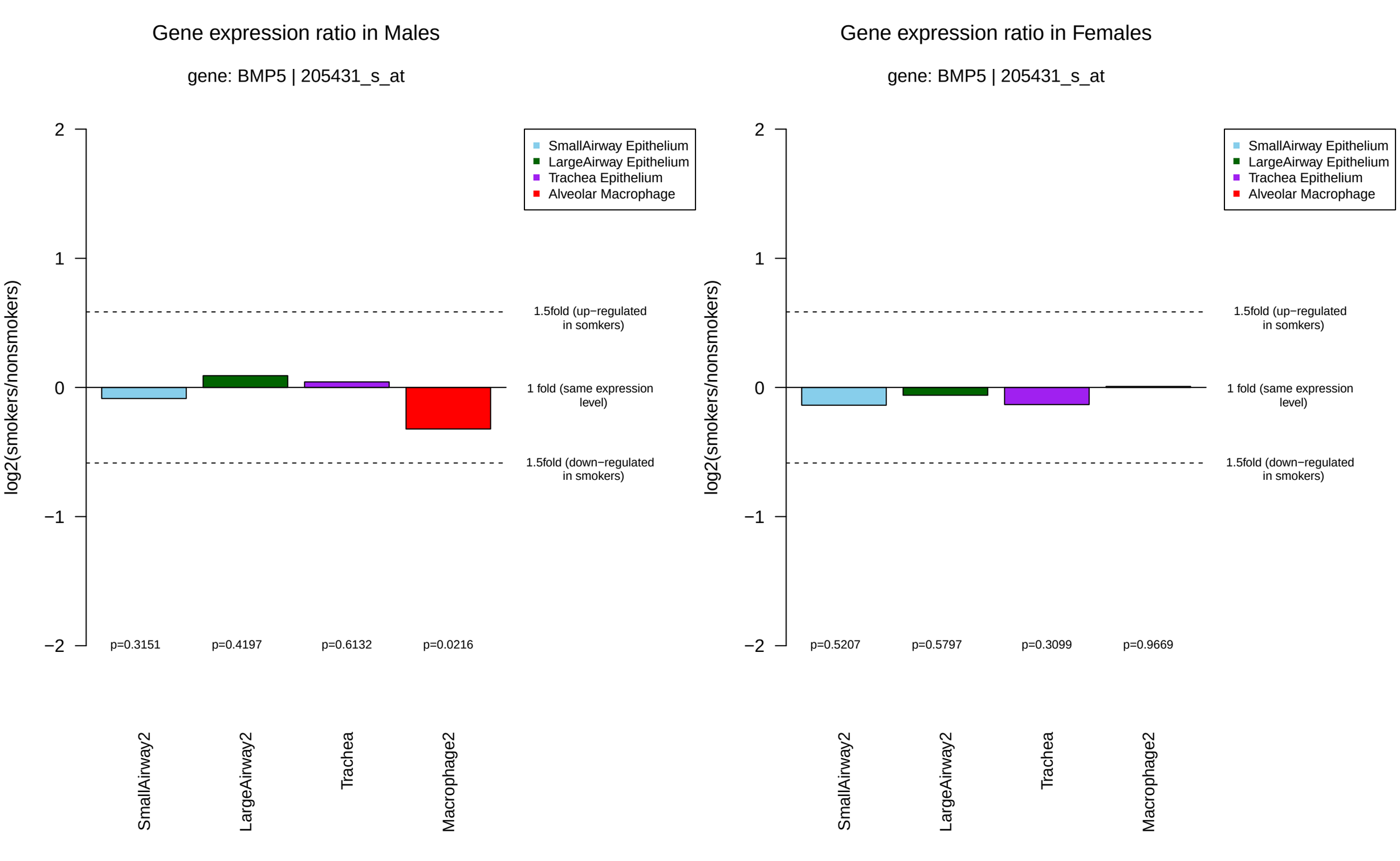
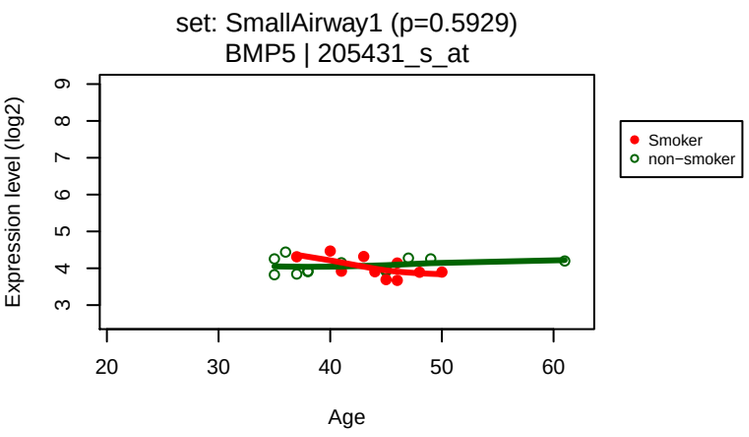
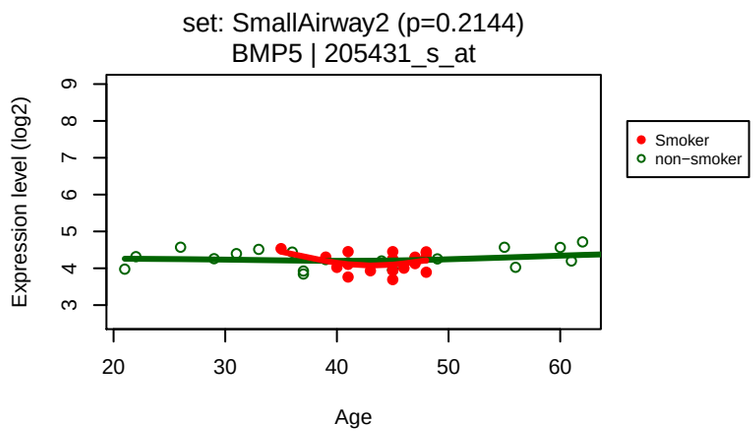
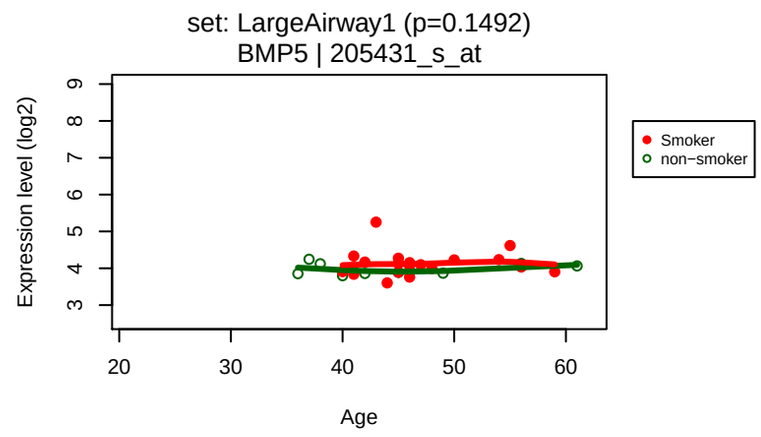

Gene expression of smokers and nonsmokers
gene: BMP5 | 205431_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5929
p=0.2144
p=0.3814
p=0.1492
p=0.5695
p=0.9543
p=0.2233
p=0.048
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

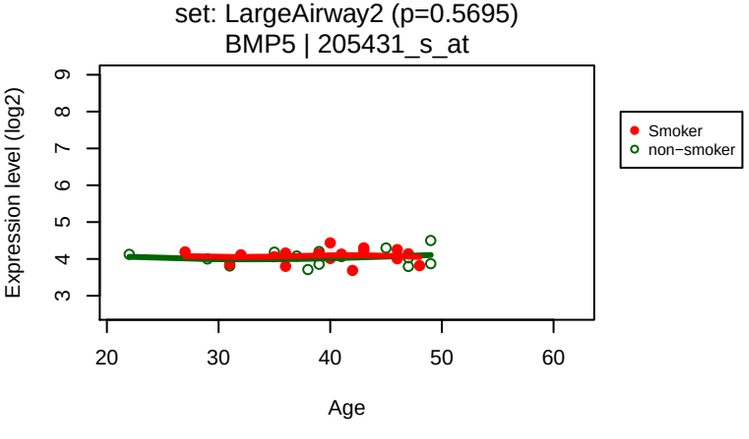

Gene expression ratio (smokers/nonsmokers)
gene: BMP5 | 205431_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5929
p=0.2144
p=0.3814
p=0.1492
p=0.5695
p=0.9543
p=0.2233
p=0.048
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

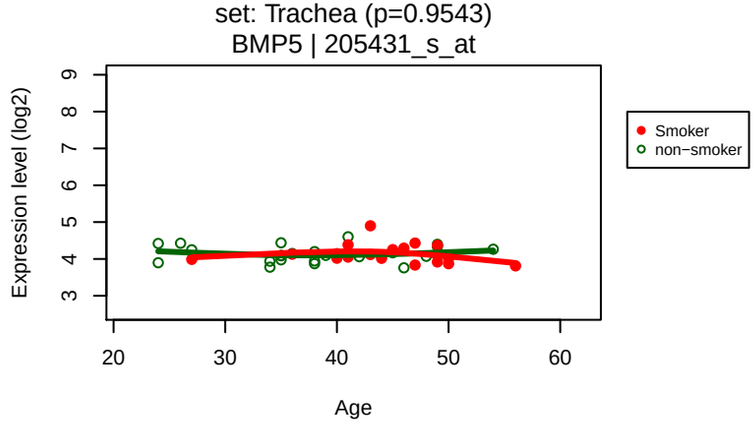

Gene expression ratio in Males
gene: BMP5 | 205431_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3151
p=0.4197
p=0.6132
p=0.0216
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMP5 | 205431_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5207
p=0.5797
p=0.3099
p=0.9669
SmallAirway2
LargeAirway2
Trachea
Macrophage2

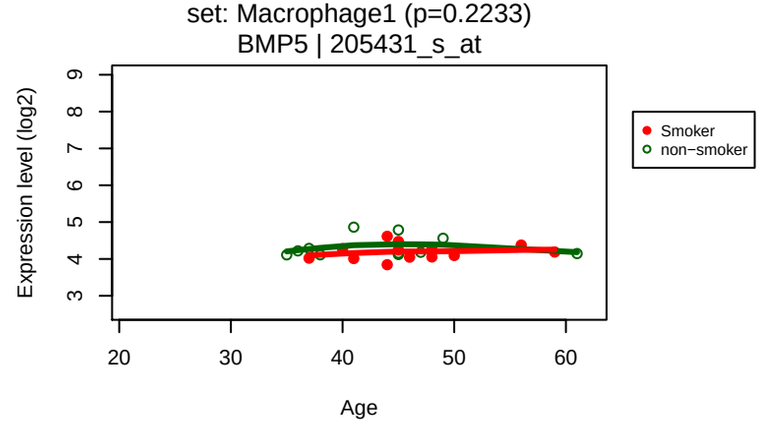

set: SmallAirway1 (p=0.5929)
BMP5 | 205431_s_at
Expression level (log2)
Age
Smoker
non−smoker

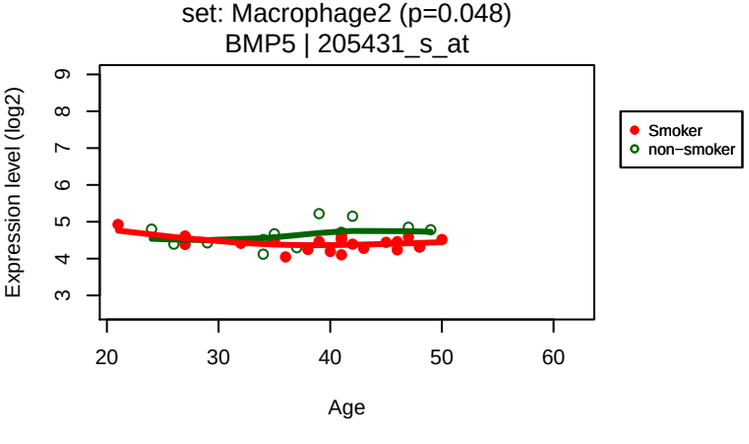

set: SmallAirway2 (p=0.2144)
BMP5 | 205431_s_at
Expression level (log2)
Age
Smoker
non−smoker

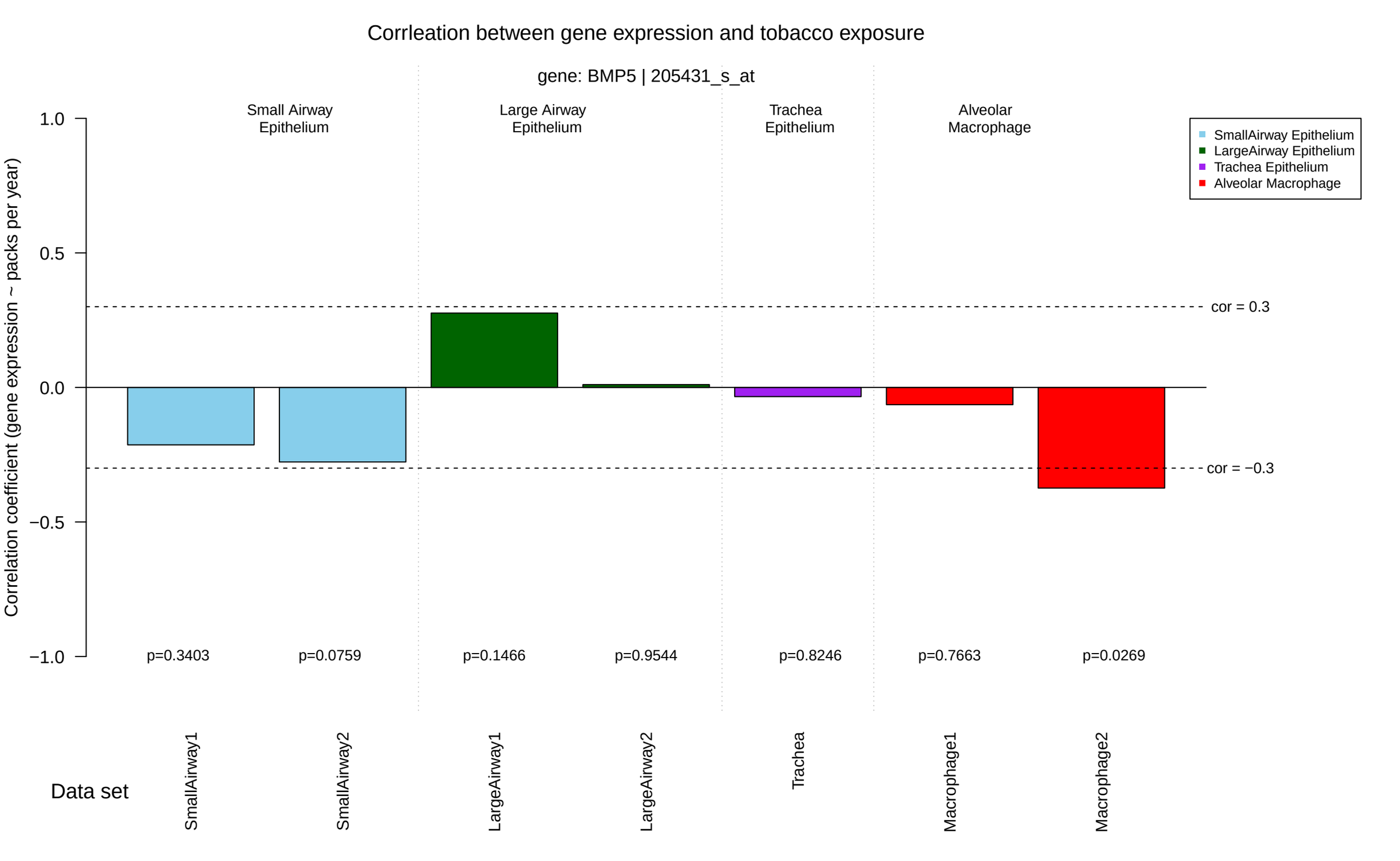

set: LargeAirway1 (p=0.1492)
BMP5 | 205431_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5695)
BMP5 | 205431_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9543)
BMP5 | 205431_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2233)
BMP5 | 205431_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.048)
BMP5 | 205431_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMP5 | 205431_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3403
p=0.0759
p=0.1466
p=0.9544
p=0.8246
p=0.7663
p=0.0269
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2